Financial Trading Strategy

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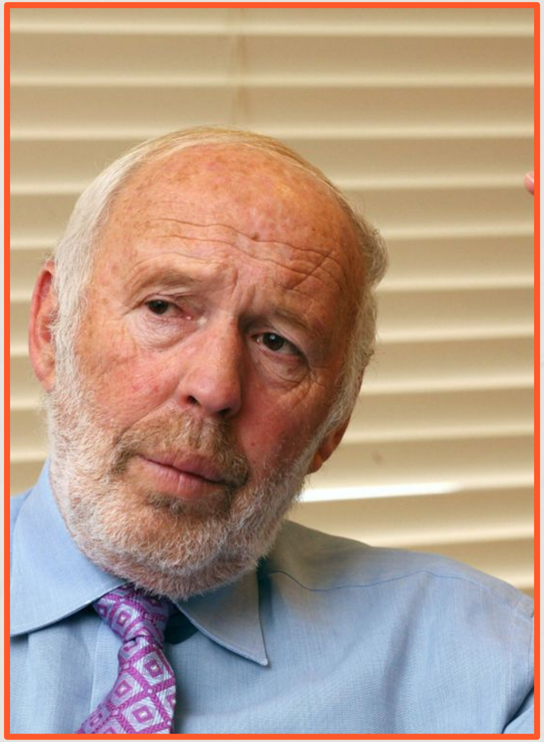


Rationale & Motivation

- Jim Simons – The man who conquered Wall Street.
- Renowned mathematician who knew nothing about finance and started a firm at age 40.
- "Patterns of price movement are not random. There are statistically significant variations that can be exploited."

Project Goals

01

Design a financial trading algorithm

02

Use real and simulated data pipelines

03

Visualize signals & evaluate risk managed trades

04

Measure performance using backtesting metrics

05

Implement machine learning

06

Learn future improvements

01

Design a financial trading algorithm

Project Roadmap - Key Initiatives

Initiative	Objective	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Onboarding	Organized 5-member team; initialize GitHub, Kanban board;												
Market Research & Strategy Design	Designed ORB strategy using 4 technical indicators												
Data Simulation, Collection, & Processing Pipeline	Processed 450K+ SPY rows from 3 years of trading data.												
Options-Based Portfolio Simulation	Simulated 700-1000 trades using Black-Scholes pricing model												
Machine Learning Integration	Trained Random Forest on engineered features.												
Performance Evaluation & Backtesting	Reformed grid search to find optimal hyperparameter combination; achieved **~7% weekly return; ~56% win rate**.												

Calls and Puts

- Calls and puts are essentially contracts that can be bought at a certain amount.
- There is an expiration date for this contract.
- There is a strike price, which is the price that the contract can be exercised at.
- To profit on calls you want new price to exceed strike price + contract cost.
- To profit on puts you want new price to be blow strike price + contract cost.

Open Range Breakout

Key Assumptions in Our Model

- Perfect fills: Calls and puts are entered at the exact Black-Scholes calculated option price with no execution lag.
- No slippage or bid/ask spread: Trades are executed at mid price with no cost incurred from spreads or market impacts
- Unlimited liquidity: You can buy/sell any number of contracts at your calculated price, without moving the market.
- Static volatility (sigma)
- Options exist at every price point: assumes there is always access to ATM calls/put contracts regardless of the price of SPY
- Portfolio growth is capped at 4x to prevent unrealistic growth.
- Trade entries are based on 1 min price intervals.
- Not inclusive of the Greeks (Gamma, Theta, Delta...)

02

Use real and simulated data pipelines

Data Sources & Generation

1-min real SPY data via yfinance API (2023–2025).

Premarket data captured separately with full day granularity.

Simulated data generated using a Black-Scholes based model.

Both pipelines are compatible with the same signal engine.

$$C = SN(d_1) - Ke^{-rt}N(d_2)$$

where:

$$d_1 = \frac{ln\frac{S}{K} + (r + \frac{\sigma_v^2}{2})t}{\sigma_s\sqrt{t}}$$

and

$$d_2 = d_1 - \sigma_s\sqrt{t}$$

and where:

$C =$ Call option price

$S =$ Current stock (or other underlying) price

$K =$ Strike price

$r =$ Risk-free interest rate

$t =$ Time to maturity

$N =$ A normal distribution

Option Pricing Calculation

I. Lanre, "Application of Black-Scholes Models for European Option Pricing," Medium, Apr. 26, 2022. [Online].

$$c = S_0\, N(d_1) - K\, e^{-rT} N(d_2)$$

$$p = K\, e^{-rT}\, N(-d_2) - S_0\, N(-d_1)$$

where $$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = \frac{\ln(S_0/K) + (r - \sigma^2/2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$

Signal Generation Logic

Open Range Breakout (ORB)

First X-min high/low of open hours of that day

BUY CALL

If Close > ORB High

BUY PUT

If Close < ORB Low

Why This Signal?

Rationale

- The market open is the most volatile time of day.
- During this period, overnight news, economic data, and investor emotion cause unpredictable price swings.
- Many people trade right at open, causing temporary volume spikes and volatility.

Our Strategy

- We define an “Open Range” based on the first 15 minutes.
- This window helps identify the high and low of the chaotic open.
- After this range is formed, price tends to settle into a direction (breakout up or down).
- We only take trades after a confirmed breakout from this range.

Why It Works

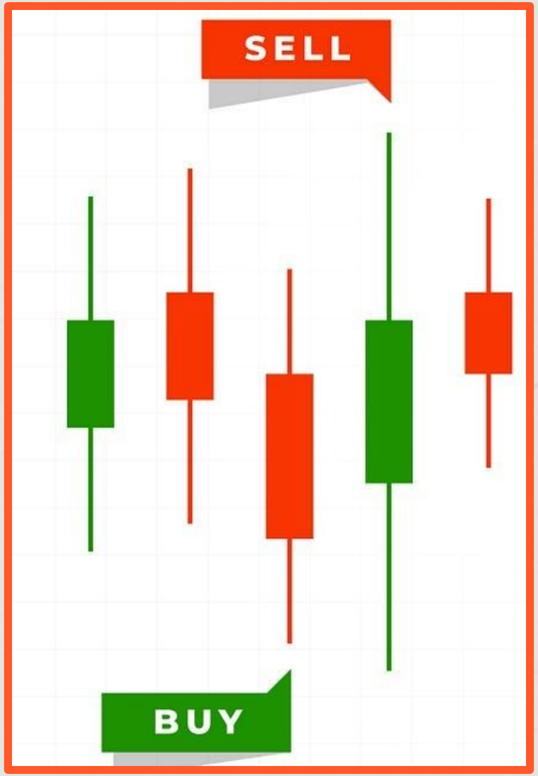
- Our algorithm waits for direction to emerge before acting.
- Reduces false signals and avoids “noise” during early market chaos.

Data Formatting

Formatted timestamps to match consistent time zone (NYC)

Added columns for: ORB High/Low

Calculated the price of option contracts during open market trading hours



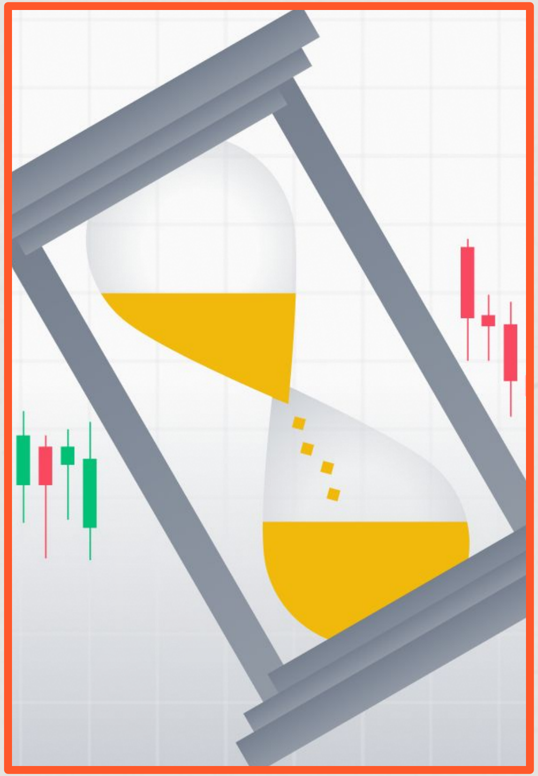
03

Visualize signals & evaluate risk managed trades

Candlestick Dashboard

Benefits of using graphs

- Plotly used for all real-time and historical visuals
- Candlestick chart with overlaid indicators (ORB, EMA, VWAP)
- Interactive: toggle indicators, zoom/pan sessions
- Pre-market and Regular session visually distinguished



04

Measure performance using backtesting metrics

Arguments in Our Back Test Function

Argument	Definition
df	Dataframe that contains price action at 1 min intervals
Initial Capital	Starting cash for this strategy
Risk per Trade	Fraction of current portfolio value to risk on each trade
Open Range Minimum	Number of minutes after market is open to define the breakout range
Stop Loss Percentage	% loss from entry options price that triggers a stop-loss exit (Static)
Take Profit Percentage	% gain from entry options that triggers a take-profit exit
Expiration Days	Number of days till expiration used to buy/price option contracts
Sigma	Assumed constant implied volatility in our model, used to price options

Arguments in Our Back Test Function

Argument	Definition
Max Trades	Maximum number of trades allowed per day to prevent over trading
Max Minutes in Trade	Maximum time a position can be held before being forced to exit
Decay Threshold	If no price moves after the max minutes in a trade and the move is less than this %, the position is sold (Adaptive)
Max Consecutive Losses	Maximum number of losing trades in a row allowed before halting trading for the day
Max Daily Loss Percentage	If total realized loss in a day exceeds this fraction of the portfolio, then trading is halted
Max Portfolio Cap Percentage	If total realized loss in a day exceeds this fraction of portfolio, then trading halts for the day
Max Drawdown Percentage	If current portfolio falls more than this %from intraday high, trading is stopped

Back Testing Performance

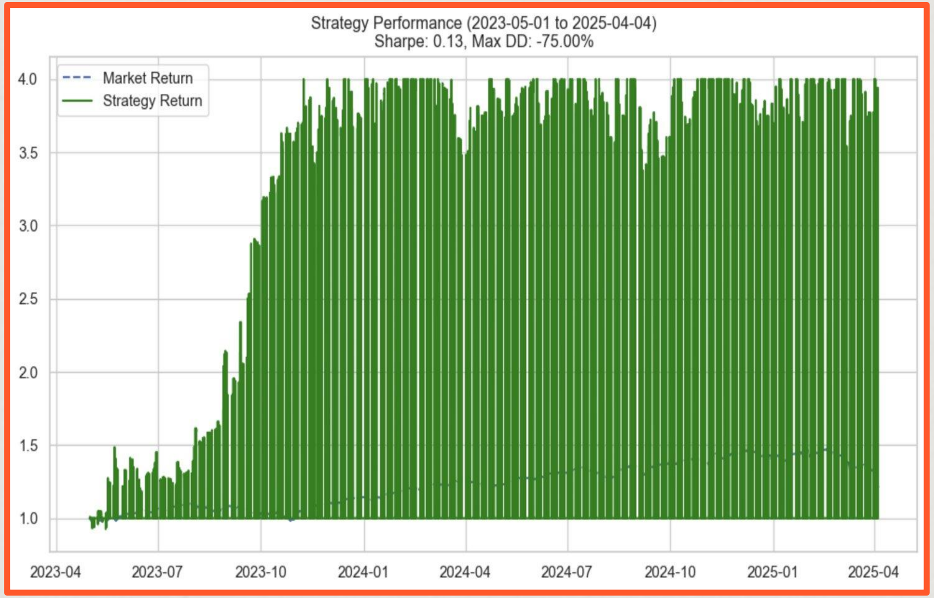
- With our simplification and assumptions, obviously the model performance will be overfitted/inaccurate
- With each strategy execution, the following values are given as outputs:

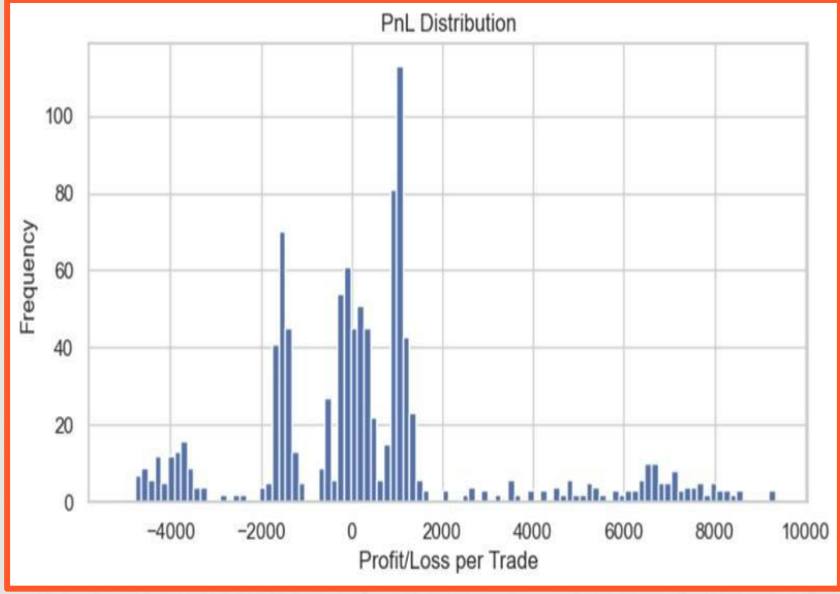
Output	Meaning
Final Portfolio Value	Ending dollar value of the portfolio after all trades executed
Cumulative Strategy Return	Total % gain\loss from start to end of strategy
Win Rate Percentage	% of trades that were profitable, where PnL > 0
Sharpe Ratio	Risk-adjusted returns; mean(daily return) / std(daily return) * sqrt(252)
Max Drawdown	Largest peak-to-trough % loss in portfolio equity
Profit Factor	Measures profit-to-risk ratio; total gross profit / total gross loss
Average Trade PnL	Mean of all profit and loss values

One Result Found with Grid Search

Parameter	Value	Parameter	Value
initial_capital	25,000	max_portfolio_cap	4.00
risk_per_trade	0.10	max_drawdown_pct	0.70
open_range_min	10.00	max_consecutive_losses	7.00
stop_loss_pct	0.15	trailing_stop_pct	0.25
take_profit_pct	0.10	decay_threshold	0.03
expiration_days	3.00	Sharpe Ratio	0.13
sigma	0.20	Max Drawdown	-75.00
max_trades	3.00	Win Rate	56.33
max_minutes_in_tra de	60.00	Final Portfolio Value	98495.54
max_daily_lass_pct	0.15	Final Strategy Return	100.00
		Total Trades	1059.00

- 0.13 Sharpe Ratio – Weak, strategy generates very little return per unit of risk
- Max Drawdown –75%, unacceptable in professional environments
- Win Rate 56.3%, over half of traders were profitable but does not ensure a strategy could win in long run if risk is not assessed properly





Graphical PnL Output

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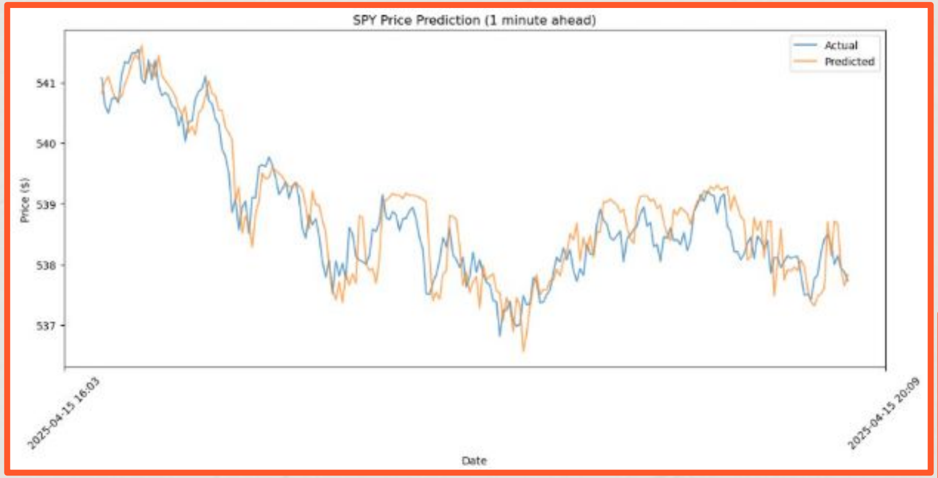
Machine Learning Implementation

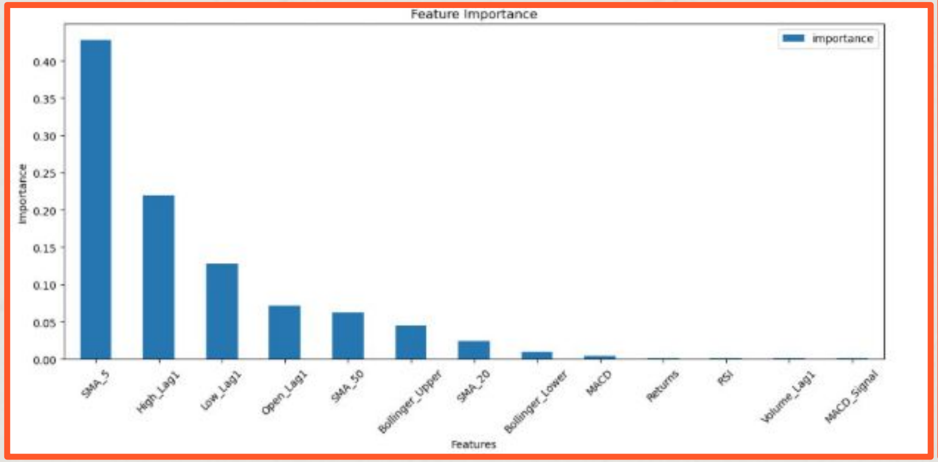
Machine Learning Module

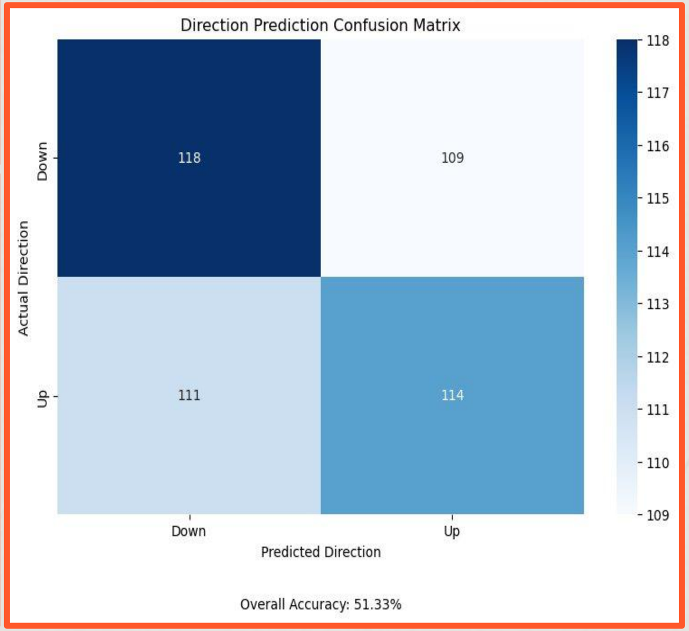
First Model: Random-Forest Price Indicator

- Random-Forest model to predict the closing price of the next minute with R^2 score of 0.83.
- Features: Returns, SMAs (5, 20, 50), RSI, MACD (and signal), and Bollinger Bands strictly using past data.
- Data: Fetches 1-minute SPY price and volume data over the last 5 days using yfinance.
- Assesses its accuracy (both in magnitude and direction) before using it to make real-time forecasts.

Outputs of Random Forest 1 Minute Predictor







Issues & Resolutions

Issues

- The first draft of the model gave results that seemed to be extremely accurate: Predicted price direction with 90% accuracy and back testing results were always profitable.
- These results were caused by the following issues with the model:
  - Data Leakage from type of data splitting
  - Lookahead bias from the features that used rolling calculations
  - RSI, MACD, and Bollinger band indicators did not use data that is strictly historical.

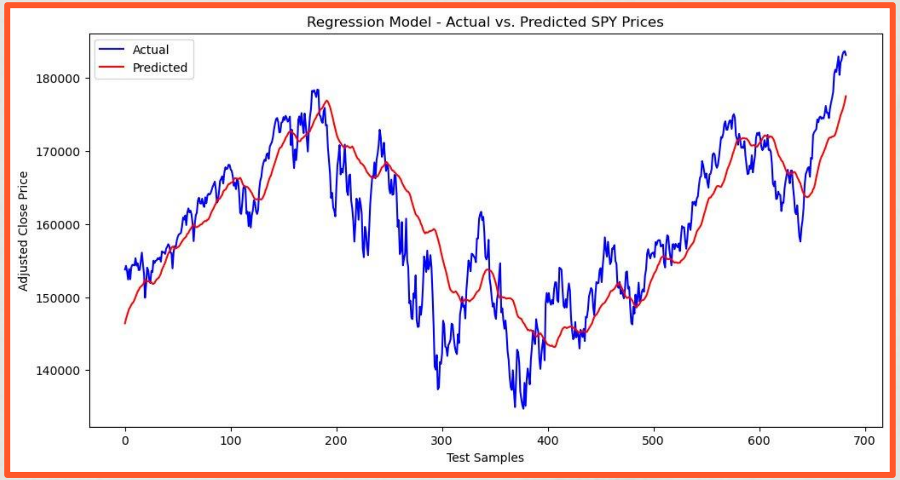
Resolutions

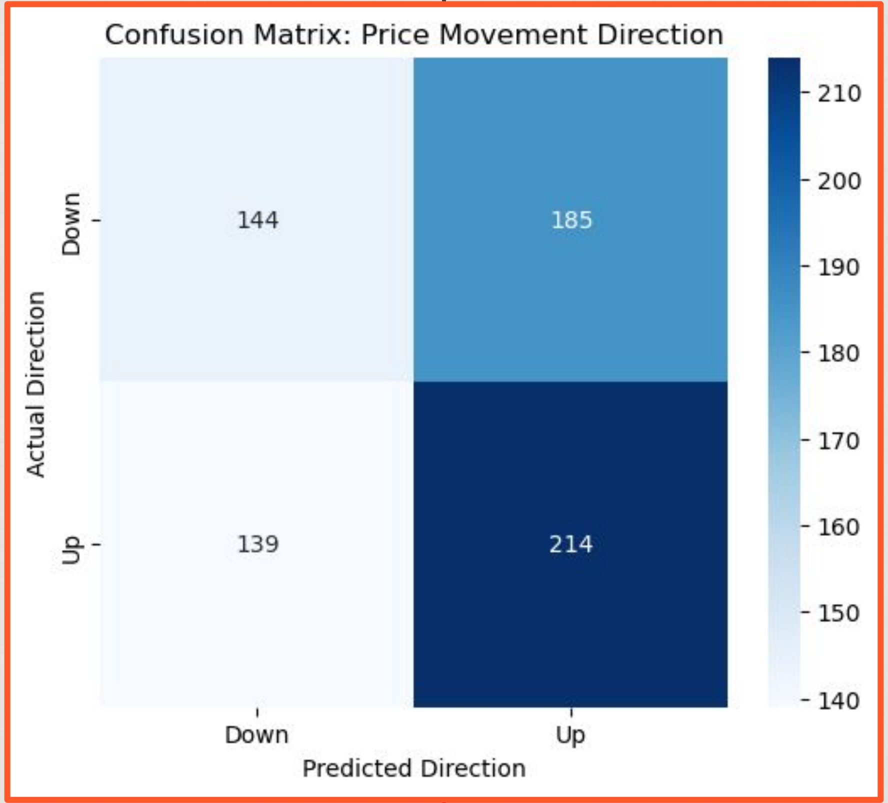
- Ensure that all technical indicators computed are based only on past values to avoid lookahead bias
- Created lagged versions of raw features to ensure that when making a prediction for a given minute, the model only uses data from previous minutes.
- The resolved code explicitly splits the dataset based on time by taking the first 80% of the data for training and the last 20% for testing. This preserves the chronological order of the data.

Machine Learning Module

Second Model: Neural Network Price Indicator

- Deep neural network regression model used to predict closing price of next day with R^2 value of 0.73 and predicts the direction of the price with 53% accuracy.
- Features: Returns, SMAs (10, 50), EMA 10, RSI
- Data: Downloads daily SPY price/volume from 2010-2023 using yfinance




Confusion Matrix: Price Movement Direction
Actual Direction
Down
Up
144
185
139
214
Down
Up
Predicted Direction
210
200
190
180
170
160
150
140

06

Future Improvements

Final Thoughts

Conclusions

- We built a fully robust simulation framework to test options-based strategies.
- Once real data is available, we can transition into production grade research.
- We plan on implementing Reinforcement Learning with other indicators to optimize the strategy; Stochastic states could include all the features and rewards.
- Greeks-Based Risk Management and signal design
- Support for real + simulated data, dynamic stop rules, and ML integration
- Include Monte Carlo and permutation testing to ensure our data is not overfitting on the current market dataset.
- Include Walk-Forward optimization and back test architecture to generalize to wider data and future market conditions.

Future Work

Real Data Integration: Source real options and stock data Clean and integrate into model

Feature Engineering and Signal Expansion: Extract and engineer features, implement new signals/indicators

Reinforcement Learning: Implement machine learning model fitting

Robustness Validation: Run Monte Carlo tests, permutation tests, walk-forward backtests

Strategy Diagnostics: Plot cumulative returns, drawdown plots, and model diagnostics

Refinement and Stress Testing: Refine model; stress test under adverse market conditions

Risk Modeling & Scenario Analysis: Quantify VaR, CVaR; simulate tail-risk events

Deployment Preparation: Paper trade with simulated funds; assess real-time behavior

Thank You!

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